HRI Group Report

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Introduction

Background

This project focuses on the Early Years Foundation Stage (EYFS), specifically on improving the experience of education for both the children and the teachers using the NAO Robot. In early education environments, teachers face increasing challenges regarding workload and “time poverty”, which limits their ability to provide personalized attention to every student (Creagh et al., 2025). This is particularly important for repetitive tasks such as vocabulary, phonics and numeric, which require consistent, one-on-one interaction to be effective.

The integration of Humanoid Artificial Intelligence Robots is becoming a more prominent solution to this issue (Lee et al., 2025). Social robots like the NAO have been used in a multitude of educational environments due to their ability to handle repetitive learning activities (Grimaldi et al., 2024). Current trends suggest that they can significantly enhance children around the Foundation stage age group in collaboration, communication, and critical thinking skills (Sapounidis, Rapti and Vaiopoulou, 2025).

Motivation

The primary motivation for this project is to explore how a humanoid robot can support early years of education through interactive, play-based learning. In this developmental stage, children learn most effectively through engagement and repetition, yet teachers often face time constraints that limit individual interaction (Creagh et al., 2025). Our goal is to use the NAO robot to assist the structured learning activities without replacing teachers.

We chose this topic because research highlights the effectiveness of play-based methods in early childhood (Bratton and Ray, 2000). The NAO robot is appropriately suited for this due to its physical appearance and expressive abilities, which can help gain attention more effectively through its social presence (Toh et al., 2016). The robot also acts as a non-judgemental partner, which can help reduce anxiety for children who may fear making mistakes in front of an adult.

While existing research may focus on complex AI systems, there is a gap in demonstrating simple and realistic classroom applications. This project addresses that gap by designing a robust "Teaching Assistant" that reinforces core concepts like numeracy and rhymes, allowing teachers to focus on higher-level tasks while the robot handles repetitive guided learning (Grimaldi et al., 2024).

Scope of Project

The project scope was defined as using the NAO robot as an interactive teaching assistant for early years learners, specifically toddlers to Year 1. The learning focus is aligned with the Early Years Foundation Stage curriculum, with particular emphasis on literacy, mathematics, understanding the world, and expressive arts and design. Within these areas, the system supports activities such as alphabet recognition, basic numeracy, shape identification, animal sounds, spelling, and simple interactive games such as “fill in the blank” and flashcards.

Several assumptions were made during development. It was assumed that interactions would take place in a controlled classroom or tutorial environment, with manageable background noise and stable lighting. The system also assumes that children interact with the robot one at a time and remain within its field of view to support face tracking and engagement. Due to the nature of toddler speech, the robot is designed to recognise a limited vocabulary and rely on simplified prompts.

Certain features were consciously excluded from the project. Advanced natural language processing, free-form conversation, and teaching children to write text were considered beyond the scope of the module due to technical complexity and time constraints. While physical pose detection and more advanced vision systems were explored conceptually, only basic tracking and engagement behaviours were prioritised to maintain system reliability.

Related Work

The use of humanoid robots in early childhood education has been widely explored in existing literature, particularly in relation to engagement, play-based learning, and repetitive instructional tasks. Educational research highlights the importance of interactive and play-oriented approaches in early development, with play therapy shown to support cognitive, emotional, and social growth in young children (Bratton and Ray, 2000).

Several studies have examined the role of robots in supporting learning activities for young children. Toh et al. (2016) provide an overview of how educational robots can be used to reinforce learning through repetition and structured interaction. This work supports the use of predictable, task-based activities within early years settings, which informed the design of our interaction flow.

More recent research focuses on the integration of humanoid robots into classroom environments. Lee et al. (2025) discuss how humanoid AI robots can assist teachers by supporting routine educational tasks rather than replacing human instruction. This perspective closely aligns with the intent of our project, which positions the NAO robot as a supportive teaching assistant.

The NAO robot has also been used as an intelligent tutoring platform in primary education. Grimaldi et al. (2024) demonstrate that NAO can maintain engagement while delivering structured learning content, influencing our decision to implement short dialogue sequences and interactive feedback loops suitable for young learners.

Additional studies suggest that educational robotics can positively impact communication, collaboration, and engagement when appropriately designed for the target age group (Sapounidis, Rapti and Vaiopoulou, 2025). Ethical and inclusivity considerations are also emphasised, particularly

the need for age-appropriate, transparent, and supportive interaction design (Tarrés-Puertas et al., 2023).

Overall, existing literature supports the feasibility of using humanoid robots as supportive tools in early years education. These studies informed the design choices made in this project and helped contextualize the implementation within the broader field of Human–Robot Interaction.

Technical Implementation

Interaction States

To manage the multi-activity educational session, we designed a hierarchical Finite State Machine (FSM). This structure allows the robot to recognise and authenticate the user before branching into different modules through facial recognition. Thus, providing a more personalised experience needed within a classroom setting.

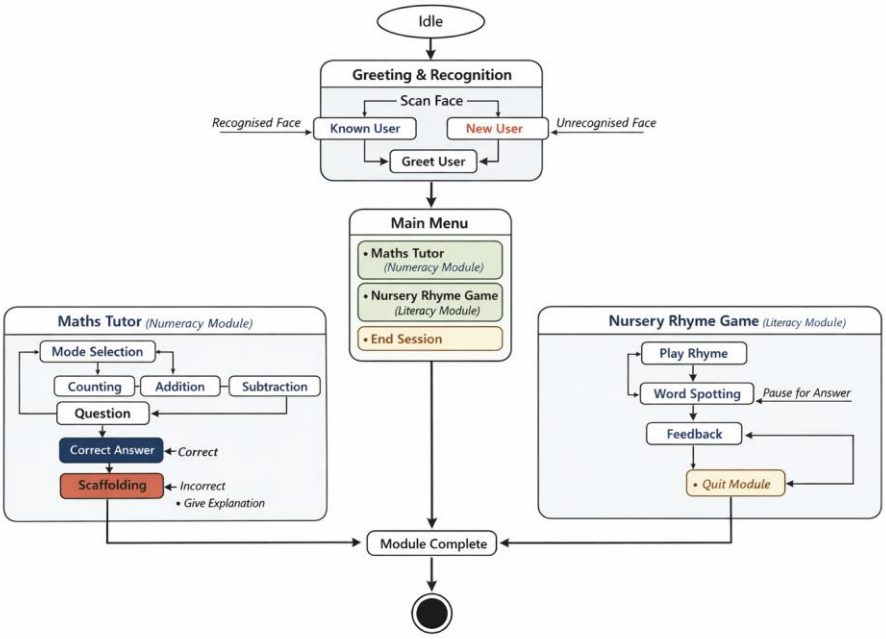


Figure 1 - Machine State Diagram showing the flow from recognition to specific modules

The interaction follows a structured flow:

1. Greeting and Recognition (State 1) The robot uses Facial detection to scan for a face.
   - Known User: If the users face is recognised, it greets the user by name.
   - New User: If the face is unrecognised, it starts a “Sign-up” state to register the new users name and map it to their FaceID (as detailed in Login logic).
2. The Maths Tutor (State 2 - Numeracy Module) This module uses a custom Python script to provide a dynamic quiz environment.
   - Mode Selection: The user can choose between the modes: Counting, Addition, Subtraction, Multiplication, or Division.

- Question Generation: The robot generates unique questions using Python's 'random' library ensuring that the child never has the same quiz twice, which helps prevent rote memorisation.
- Scaffolding technique: If the user answers incorrectly, the system does not just say "Wrong." It causes an explain_answer() method that provides a contextual explanation (e.g., "Division is sharing. If you share 4 cookies between 2 people..."), which aligns with constructive teaching methods.

3. The Nursery Rhyme Game (State 3 - Literacy Module):
   - The Robot plays a rhyme and uses "Word Spotting" to pause and wait for the user to fill in the missing word (e.g., "Twinkle twinkle little..." -> "Star"). This promotes active listening and vocabulary recall for the child.

Dialogue Capabilities

The system uses two different methods for handling dialogue, depending on the complexity of the task.

Standard Dialogue (State 1 and State 3)

For the Nursery rhyme section and Greeting section, we used the standard Choreographe ALTextToSpeech boxes. This allowed us to easily coordinate the robot's gestures with its speech using the built-in timeline, ensuring smooth and animated performance for the songs and facial recognition.

Python-Controlled Dialogue (State 2)

For the Maths Tutor, we need the robot to speak dynamically generated numbers and explanations. However, we realised that the standard speech commands would conflict with the robot's listening loop in Python.

To solve this, we implemented a custom helper function, say_statue(), Specifically for this Python script. This function sets the robot's bodyLanguageMode to "disabled" during the quiz questions. This ensures the robot remains focused and stable while calculating questions, preventing any motor interference during the processing of the logic.

External Systems

Face Recognition system:

We used the NAOs built-in ALFaceDetection module to map a unique FaceID to a stored username string. This allows the robot to maintain a long-term memory of each student, greeting them personally when a new session is started.

Word Spotting for Speech Recognition:

We bypassed cloud-based recognition for local "Word Spotting" in the Maths module. The vocabulary list is updated to listen only for the relevant numbers based on the context. For example, when the answer is 4, it prioritises "Four". This helps to reduce false positives and negatives from background noise compared to an open-ended speech engine.

Code Overview

The counting and maths logic required a custom Python class (MyClass) to handle the randomisation and feedback.

```
elif self.mode == 'div':
    answer = random.randint(1, 4)
    divisor = 2
    dividend = answer * divisor
    self.correct_answer = answer
    self.num_a = dividend
    self.num_b = divisor
    self.say_statue("What is {} divided by {}?".format(dividend, divisor))
```

Randomized Question Logic:

To prevent repetitive learning, the Python script generates questions dynamically using the 'random' library. For example, the Division logic ensures only integer answers are generated by reversing the multiplication operation:

Contextual Feedback:

The Script also handles educational "scaffolding". Instead of a generic "Wrong" message, the explain_answer method constructs a specific hint based on the math mode selected:

```
def explain_answer(self):
    self.say_statue("Not quite. The answer is {}.".format(self.correct_answer))

    if self.mode == 'count':
        self.say_statue("We are just counting up by ones. After {}, comes {}.".format(self.correct_answer - 1, self.correct_answer))
    elif self.mode == 'add':
        self.say_statue("If you put {} fingers and {} fingers together, you get {}.".format(self.num_a, self.num_b, self.correct_answer))
    elif self.mode == 'sub':
        self.say_statue("Subtraction means taking away. If you have {} and take away {}, only {} are left.".format(self.num_a, self.num_b, self.correct_answer))
    elif self.mode == 'mult':
        self.say_statue("Multiplication is adding groups. {} groups of {} is {}.".format(self.num_a, self.num_b, self.correct_answer))
    elif self.mode == 'div':
        self.say_statue("Division is sharing. If you share {} cookies between {} people, everyone gets {}.".format(self.num_a, self.num_b, self.correct_answer))
```

The hybrid approach of using standard Choreographe boxes for the performance elements and Python for the logic-heavy parts allowed us to utilize the NAOs capabilities to its best strengths.

HRI Dialogue Example

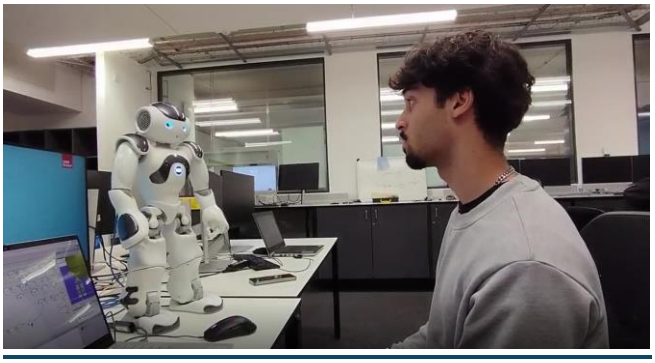


Figure 2 - Face Recognition function



Figure 3 - teaching "Counting Numbers" Function

```
NAO Robot: Hello, My name is NAO.
NAO Robot: Please stand in front of me so I can see your face.
NAO Robot: Please hold still.
NAO Robot: Your name is Nihao.
NAO Robot: Is that correct?
Student  : Yes.
NAO Robot: It's good to see you.
NAO Robot: What would you like to do today?
Student  : Numbers.
NAO Robot: That sounds great.
NAO Robot: I love numbers.
NAO Robot: How about this?
NAO Robot: I will say the numbers from 1:00 to 10:00.
NAO Robot: Then you will repeat after each one.
NAO Robot: Does that sound good?
Student  : Yes.
NAO Robot: OK, Say ready when you are ready.
Student  : Ready
NAO Robot: 1
Student  : 1
NAO Robot: Good
(For simplicity 2-9 was skipped)
NAO Robot: 10
Student  : 10
NAO Robot: Great job.
NAO Robot: Now, what would you like to do today?
```

Figure 4 - Nao Robot Number Function Dialogue

Face Tracking and Engagement Initialization

Before the educational task begins, the robot ensures the environment meets the project's assumptions of a controlled, one-on-one interaction. The robot prompts the student to "stand in front of me so I can see your face" and "hold still". This establishes the face-tracking link required to maintain engagement during the lesson.

A primary technical challenge identified in the project is the limitation of the NAO's speech recognition, especially with toddler speech patterns or non-generic names.

- **Error Diagnosis**: Rather than assuming the name was captured correctly, the robot utilizes a Verification Loop by repeating the name it heard ("Your name is Nihao. Is that correct?").
- **Recovery Strategy**: If the user says "No," the system is designed to loop back and ask for the name again until a correct match is confirmed. While this can lead to user frustration if repeated multiple times, it is a necessary robust design choice to ensure accuracy before proceeding.

- **Performance Observation**: The robot tends to succeed on the first attempt with generic names but struggles with more unique phonetic variations.

Once the social "handshake" (greeting and naming) is complete, the robot transitions into its role as a Teaching Assistant.

- **Simplified Prompts**: The robot uses closed-ended questions ("Does that sound good?") and clear instructions ("Say ready when you are ready") to guide the child through the activity.
- **Repetitive Reinforcement**: The counting exercise (1 through 10) directly supports the mathematics curriculum through guided repetition.
- **Positive Feedback**: By saying "Good" and "Great job," the robot acts as the "non-judgmental partner," providing immediate social rewards that encourage the student to continue without fear of making a mistake.

Discussion

Plan vs Implementation

Our initial project plan was aimed at creating a multi-modal “Teaching Assistant” capable of easily switching between subjects (Maths, Literacy, Games) based on natural language requests from the child. We also intended to add physical object recognition like identifying flashcards of fruits/animals on top of the facial recognition.

However, during the implementation stage these plans were deviated from. While we successfully developed individual modules such as the Python-based maths tutor, the Choreographe based rhyme game, the facial recognition system and a counting module. Compiling them into a single system that worked together was a lot more difficult than originally expected.

Compilation challenges

The primary technical reason for the deviation was the challenge of combining the different coding architectures. The maths module relied on a custom python class with specific thread blocking (to wait for answers), whereas the rhyme module used Choreographe flowcharts. Attempting to compile these into a single script resulted in conflict errors where the robot would fail to exit one mode and enter the next cleanly.

Project management

Reflecting on the project, the team underestimated the time required for this system integration the technical differences raised from the robot compared to its simulation model. We allocated most of our time into developing features separately rather than testing the integrated switching logic. This led to a forced descope of the final presentation, to just include what we were able to integrate and have showtime ready.

Limitations

Despite some of the individual modules working as intended, a multitude of limitations with our final project are still there.

Speech recognition and natural language:

The most significant limitation is the reliance on “Word Spotting” instead of having true Natural Language Processing. The robot expects to hear specific keywords, if a child uses a synonym or speaks irregularly the system may fail to work. Toddlers and younger children tend to have more variance in their speech pattern and natural language which may not be understood correctly by the NAO robot. While we were able to mitigate this by lowering the confidence threshold, this increases the likelihood of false positives. A better solution would require offloading the audio processing to a cloud service or to an LLM.

Latency and engagement:

When using the NAO robot, we frequently ran into latency issues with the robots replies to the user. This particularly happened when using the maths module caused by the processing time that comes with using Python. Maintaining the attention of the children is essential especially in early years education and the slow response may break it. A future iteration of our robot could implement filler animations to keep the child engaged whilst the robot is processing speech.

Conclusion

This project successfully demonstrates the viability of the NAO robot as a robust "Teaching Assistant" tailored for the Early Years Foundation Stage (EYFS**)**. By addressing the critical issue of "time poverty" among early education teachers, the system provides an effective solution for managing repetitive but essential tasks such as numeracy, phonics, and vocabulary reinforcement.

Key Findings & Outcomes

- **Social Presence and Engagement**: The robot’s physical embodiment and expressive abilities act as a "non-judgmental partner," which significantly reduces anxiety for young learners (Toddlers to Year 1) and encourages persistent effort without fear of mistakes.
- **Technical Robustness**: Despite the inherent challenges of toddler speech recognition, the implementation of a Verification Loop—where the robot repeats the name it heard for confirmation—provides a reliable recovery strategy that maintains the flow of interaction.
- **Curriculum Alignment**: The system effectively supports core EYFS pillars, including literacy and mathematics, through structured, play-based activities like "fill in the blank" and flashcard recognition.

Stakeholders and Impact

The results of this project are particularly relevant to EYFS educators and school leaders seeking to integrate AI tools that augment rather than replace human instruction. By acting as a non-judgmental partner, the robot helps reduce student "mistake anxiety," creating a supportive environment for foundational learning

Future Outlook

While the current scope prioritized system reliability by excluding complex natural language processing, future iterations could benefit from more advanced "word spotting" or vision-based gesture

recognition to further enrich the interaction. Ultimately, this project reinforces that humanoid robots are most effective when used to augment rather than replace the teacher, allowing human educators to focus on higher-level developmental tasks while the robot handles the guided repetition necessary for foundational learning.

References

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Appendices

Videos used in presentation:

hriNaoFaceRecoExample.mp4

hriNaoRun.mp4

GitHub repository:

https://github.com/mkwill12/UWE-HRI-Group-Project